Harshwardhan Bhosale

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Summary

Aspiring full-stack web developer with hands-on experience in React, HTML, CSS and JavaScript, and in building REST APIs with JWT authentication. Currently deepening backend skills with Node.js, Express and MongoDB. A data science and LLM background (Python, scikit-learn, RAG) adds the ability to build data-driven and AI-powered web applications. Strong leadership qualities, with the ability to guide teams, take ownership and deliver results.

Skills

Languages:
JavaScript, Python, Java

Backend:
Node.js, Express, REST APIs, JWT Authentication, bcyrpt

Data Analysis & Visualization:
Pandas, Matplotlib, Seaborn, Exploratory Data Analysis

Tools & Services:
VS Code, Postman, GitHub, Vercel, Jupyter Notebook

Frontend:
React, Vite, Sass, Responsive Design, Tailwind CSS

AI & Machine Learning:
Scikit-learn, NumPy, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings & Semantic Search, OpenAI Whisper, Ollama

Database:
MySQL, Mongoose, MongoDB

Projects

AI Teacher Assistant
Video Question Answering System using Retrieval-Augmented Generation (RAG)
- Built an end-to-end RAG pipeline over 20 Java lecture videos (1.9 hours), using FFmpeg for audio extraction and OpenAI Whisper large-v2 for timestamped English transcripts (Hindi-to-English translation).
- Split transcripts into 372 timestamped chunks, generated 1024-dimensional bge-m3 embeddings, and implemented cosine-similarity search to retrieve the top 5 chunks; a local LLM (Gemma 2 via Ollama) returns the lesson number and timestamps at zero API cost.
- Evaluated on 40 synthetic questions: 97.5% top-5 lesson accuracy and 92.5% timestamp accuracy (within 60 seconds), with about 6 seconds response time on local hardware.

Productivity Dashboard
JavaScript, HTML, CSS, API, Local Storage
- Built a productivity dashboard with four modules (To Do List, Daily Planner, Motivation and Pomodoro Timer) and a header showing the live date, time and location.
- Integrated external APIs to display real-time weather (temperature, conditions, humidity, wind speed) and load fresh motivational content dynamically.
- Used Local Storage to save daily planner entries so they persist after the page is closed or refreshed, and built a Pomodoro timer for focus and break cycles.

Professional Experience

Prodigy Infotech 12/2025 – 01/2026
Data science
- Completed four data science tasks in Python covering data cleaning, EDA, visualization, and machine learning.
- Built a Decision Tree classifier (scikit-learn) to predict customer purchases from bank marketing data.
- Performed sentiment analysis on Twitter data and visualized insights using Matplotlib and Seaborn.

Education

Bachelor of Engineering 08/2023 – Present
PES Modern College of Engineering, Pune Pune, India